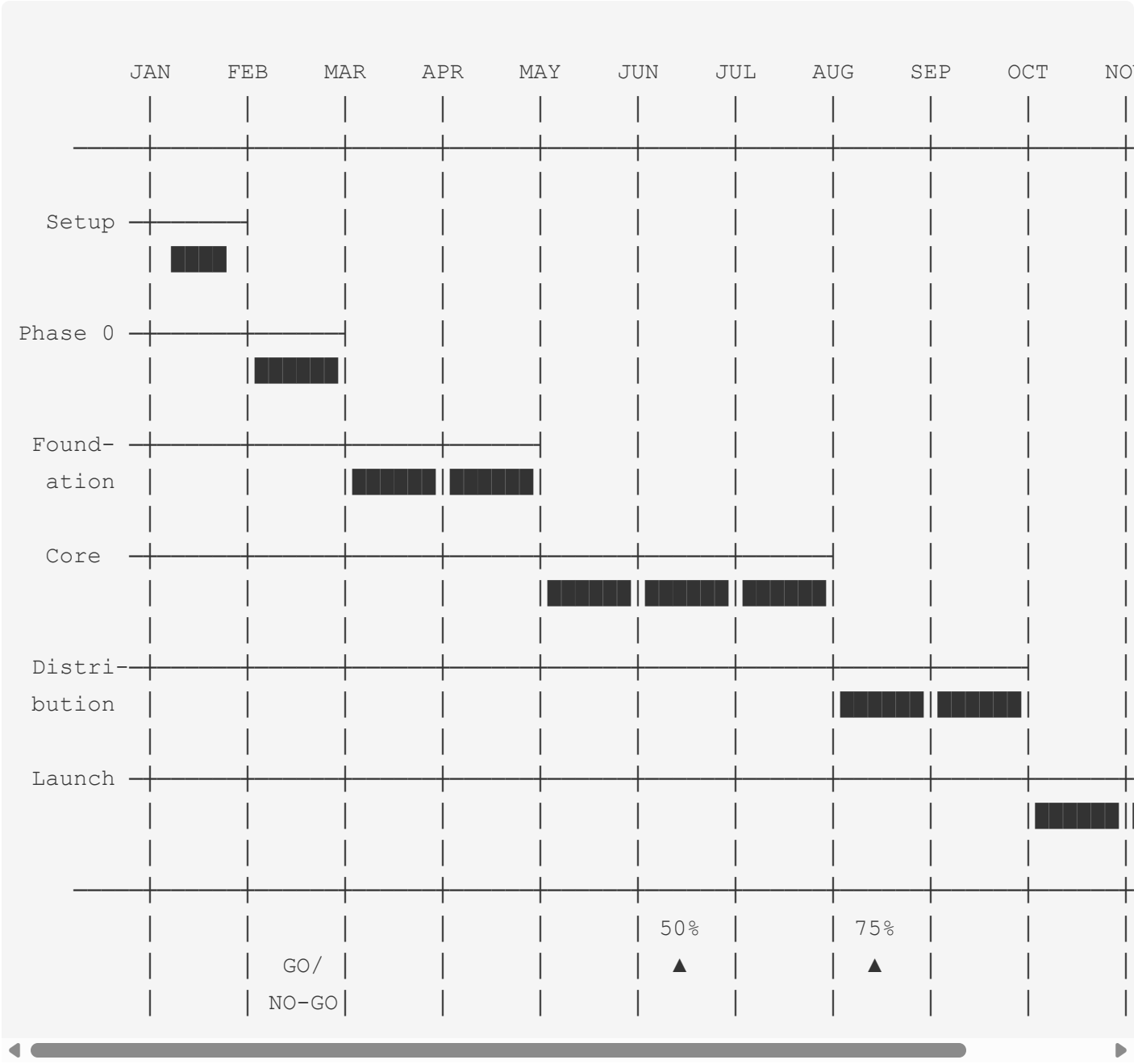
.things that Matter

Development Roadmap & Milestones

Document Information	
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Company Name	**.things that Matter**
Document Type	**Development Roadmap**
Version	**1.0**
Date	**January 21, 2026**
Author	**Matome Mboweni, CTO**

1. Roadmap Overview

1.1 Phase 1 Timeline (2026)



1.2 Sprint Structure

Sprint	Duration	Focus
Setup	2 weeks	Environment, contracts, planning

Phase 0	**4 weeks**	**Steganography validation**
Foundation	**8 weeks**	**FlatBuffers, Rust compiler**
Core	**12 weeks**	**CLIP, signing, steganography engine**
Distribution	**8 weeks**	**CDN, API, browser extension**
Launch	**8 weeks**	**Beta merchants, public release**

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2. Detailed Sprint Breakdown

2.1 Sprint 0: Setup (Jan 21 - Feb 2, 2026)

Duration: 2 weeks **Objective:** Establish foundation for development

Week	Tasks	Deliverables
---	---	---
Week 1	**Sign new company contract**	**Signed agreement**
	Set up development environment	**Cursor Pro, VS Code, Rust toolchain**
	Create GitHub organization	**thingsthatmatter org**
	Initialize monorepo structure	**packages/ directory**

Week 2	**Set up Supabase project**	**Database schema**
	Configure Vercel deployment	**CI/CD pipeline**
	Set up Cloudflare R2 bucket	**Storage ready**
	Begin Rust learning	**Chapters 1-5 of The Rust Book**

Exit Criteria: - [ ] Contract signed - [ ] Development environment operational - [ ] Repository initialized with structure - [ ] Infrastructure accounts created

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2.2 Sprint 1: Phase 0 - Validation (Feb 3 - Mar 2, 2026)

Duration: 4 weeks **Objective:** Prove steganography viability **Budget:** R2,000

Week	Tasks	Deliverables
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Week 3	**Research steganography techniques**	**Technical notes**
	Build basic XMP injection tool (Rust)	**xmp_inject.rs**
	Build EOF append tool (Rust)	**eof_append.rs**
	Test on 5 platforms	**Initial results**
Week 4	**Test on 15 additional platforms**	**Full platform matrix**
	Document survival rates	**Platform compatibility table**

	Build basic extraction tool	extract_payload.py
	CLIP API integration test	clip_test.py
Week 5	Build end-to-end prototype	JSON → .thing → image → decode
	Cost analysis for CLIP	Cost projections
	Validate Ed25519 signing	sign_verify.py
Week 6	Compile results	Phase 0 Report
	Go/No-Go decision	Decision document
	Plan adjustments if needed	Updated roadmap

Platform Testing Matrix:

Platform	Test	Status
---	---	---
Discord	Upload/download image	Pending
Telegram	Send/receive image	Pending
Twitter/X	Tweet with image	Pending
Instagram	Post image	Pending
Facebook	Share image	Pending

WhatsApp	**Send image**	**Pending**
Signal	**Send image**	**Pending**
Reddit	**Post image**	**Pending**
Pinterest	**Pin image**	**Pending**
LinkedIn	**Post image**	**Pending**
Slack	**Share image**	**Pending**
Email (Gmail)	**Attach image**	**Pending**
Email (Outlook)	**Attach image**	**Pending**
Google Photos	**Upload**	**Pending**
iCloud Photos	**Upload**	**Pending**
Dropbox	**Upload**	**Pending**
Google Drive	**Upload**	**Pending**
OneDrive	**Upload**	**Pending**
Direct download	**HTTP transfer**	**Pending**
S3/R2	**CDN delivery**	**Pending**

Decision Gate:

Survival Rate	Decision
---	---
≥70%	Full steam ahead with steganography
50-69%	Hybrid approach: steg + matter:// URIs
<50%	Pivot to matter:// as primary distribution

Exit Criteria: - [ ] 20 platforms tested - [ ] Survival rate documented - [ ] End-to-end prototype working - [ ] Go/No-Go decision made - [ ] Phase 0 Report complete

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2.3 Sprint 2-3: Foundation (Mar 3 - Apr 27, 2026)

Duration: 8 weeks **Objective:** Build compiler foundation

Sprint 2: FlatBuffers & Schema (Mar 3 - Mar 30)

Week	Tasks	Deliverables
---	---	---
Week 7	Design FlatBuffers schema	thing.fbs
	Set up flatc compiler	Build scripts
	Generate Rust bindings	thing_generated.rs
	Write schema documentation	Schema docs

Week 8	**Implement basic serializer (Rust)**	**serialize.rs**
	Implement basic deserializer (Rust)	**deserialize.rs**
	Unit tests for serialization	**tests/serialize.rs**
	Zstd compression integration	**compress.rs**

Sprint 3: Rust Compiler Foundation (Mar 31 - Apr 27)

Week	**Tasks**	**Deliverables**
------	**-------**	**--------------**
Week 9	**Product JSON validation (serde_json + validator)**	**validator.rs**
	Image download utility	**fetch_image.rs**
	CLIP API integration	**clip_client.rs**
	Vector embedding pipeline	**vectorize.rs**
Week 10	**Ed25519 signing implementation**	**crypto.rs**
	Signature verification	**verify.rs**
	Key management utilities	**keys.rs**
	Integration tests	**tests/crypto.rs**
Week 11	**Full compile() function**	**compiler.rs**

	CLI tool	**matter-cli (Rust binary)**

	Batch processing support	**batch.rs**

	Error handling & logging	**errors.rs**

Week 12	**End-to-end testing**	**tests/e2e.rs**

	Performance benchmarking	**benches/**

	Documentation	**compiler_docs.md**

	Code review & cleanup	**Refactored code**

Milestone: 30% Complete

Exit Criteria: - [ ] FlatBuffers schema finalized - [ ] Rust compiler produces valid .thing files - [ ] CLIP vectors embedded correctly - [ ] Ed25519 signatures verify correctly - [ ] CLI tool operational - [ ] <500ms compile time achieved

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2.4 Sprint 4-6: Core (Apr 28 - Jul 20, 2026)

Duration: 12 weeks **Objective:** Complete core functionality

Sprint 4: Steganography Engine (Apr 28 - May 25)

Week	Tasks	Deliverables
---	---	---
Week 13	**XMP metadata injection library**	**xmp.rs**

	EOF binary append library	**eof.rs**
	Image format handling (JPEG/PNG)	**image_utils.rs**
	Polyglot image generator	**polyglot.rs**
Week 14	**XMP beacon format specification**	**xmp_spec.md**
	Gist JSON structure	**gist.rs**
	Metadata extraction library	**extract.rs**
	Unit tests	**tests/steg.rs**
Week 15	**Integration with compiler**	**Updated compiler.rs**
	Output: .thing + polyglot image	**Full pipeline**
	Quality validation	**image_quality.rs**
	Documentation	**steg_docs.md**
Week 16	**Platform-specific optimizations**	**platform_compat.rs**
	Fallback strategies	**fallback.rs**
	Error recovery	**recovery.rs**
	Performance tuning	**Optimized code**

Sprint 5: Shadow Index API (May 26 - Jun 22)

Week	Tasks	Deliverables
------	-------	--------------
Week 17	API design & OpenAPI spec	openapi.yaml
	Supabase schema (hash → metadata)	migrations/
	POST /compile endpoint	api/compile.ts
	GET /resolve endpoint	api/resolve.ts
Week 18	GET /gist endpoint	api/gist.ts
	POST /batch endpoint	api/batch.ts
	Authentication (API keys)	auth.ts
	Rate limiting	ratelimit.ts
Week 19	Cloudflare R2 integration	storage.ts
	Upload pipeline	upload.ts
	CDN URL generation	cdn.ts
	Caching layer (Redis)	cache.ts
Week 20	Error handling	errors.ts

	Logging & monitoring	**logging.ts**
	API documentation	**api_docs.md**
	Deploy to Vercel	**Production API**

Milestone: 50% Complete

Sprint 6: matter:// Protocol (Jun 23 - Jul 20)

Week	**Tasks**	**Deliverables**
---	---	---
Week 21	**URI scheme specification**	**matter*uri*spec.md**
	Hash encoding (Base58)	**encoding.ts**
	URI generation in compiler	**Updated compiler**
	URI parsing library	**uri_parser.ts**
Week 22	**Protocol handler research**	**Browser compatibility notes**
	Fallback HTTPS URLs	**fallback_urls.ts**
	Deep linking support	**deeplink.ts**
	Documentation	**protocol_docs.md**
Week 23	**Rust: Steganography integration**	**matter-core/src/steg/**

	Rust: CLIP client optimization	**matter-core/src/clip/**
	WASM build optimization	**Smaller WASM**
	Performance benchmarks	**benches/**
Week 24	**Rust: Full pipeline optimization**	**matter-core/src/compiler.rs**
	Integration tests	**tests/integration.rs**
	Documentation	**rust_docs.md**
	Code review & refactoring	**Optimized codebase**

Exit Criteria: - [ ] Steganography engine complete - [ ] Shadow Index API deployed - [ ] matter:// URI generation working - [ ] Rust port 30% complete - [ ] All core features functional

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2.5 Sprint 7-8: Distribution (Jul 21 - Sep 14, 2026)

Duration: 8 weeks **Objective:** Browser extension and distribution infrastructure

Sprint 7: Browser Extension (Jul 21 - Aug 17)

Week	**Tasks**	**Deliverables**
------	**-------**	**----------**
Week 25	**Extension architecture design**	**Architecture doc**
	Manifest v3 setup	**manifest.json**

	Content script: DOM observer	**content.js**
	XMP beacon detection	**detect.js**
Week 26	**Background worker: fetch .thing**	**background.js**
	WASM: libmatter compilation	**libmatter.wasm**
	Signature verification in WASM	**verify.wasm**
	IndexedDB caching	**cache.js**
Week 27	**Overlay UI: Shadow DOM**	**overlay.js**
	Product card component	**card.js**
	Styling (isolated CSS)	**styles.css**
	"Buy Now" flow	**checkout.js**
Week 28	**matter:// URI interception**	**protocol.js**
	Settings page	**settings.html**
	Error handling	**errors.js**
	Testing & debugging	**test_extension.js**

Milestone: 75% Complete

Sprint 8: Polish & Integration (Aug 18 - Sep 14)

Week	Tasks	Deliverables
------	-------	--------------
Week 29	Extension performance optimization	Optimized code
	Bundle size reduction	<100KB target
	Cross-browser testing (Chrome, Edge)	Compatibility report
	Accessibility review	A11y fixes
Week 30	Integration: compiler + API + extension	Full pipeline test
	End-to-end testing	E2E test suite
	Bug fixes	Issue resolution
	Documentation	User guide
Week 31	Rust port: steganography	steg.rs
	Rust port: CLIP client	clip.rs
	WASM build optimization	Smaller WASM
	Performance benchmarks	Benchmark report
Week 32	Security review (self)	Security checklist

	Code cleanup	**Refactored code**

	README updates	**Documentation**

	Prepare for beta	**Beta checklist**

Exit Criteria: - [ ] Browser extension functional - [ ] End-to-end demo working - [ ] Rust compiler 50-70% complete - [ ] WASM verification working - [ ] Ready for beta testing

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2.6 Sprint 9-10: Launch (Sep 15 - Dec 15, 2026)

Duration: 13 weeks **Objective:** Beta launch and public release

Sprint 9: Beta Program (Sep 15 - Oct 26)

Week	Tasks	Deliverables
---	---	---
Week 33	**Beta merchant onboarding (5-10)**	**Merchant accounts**
	Merchant documentation	**Onboarding guide**
	Support process setup	**Support workflow**
	Feedback collection system	**Feedback forms**
Week 34	**Compile merchant catalogs**	**1000+ products**
	Monitor performance	**Metrics dashboard**

	Bug triage & fixes	**Bug fixes**
	Merchant feedback calls	**Feedback notes**
Week 35	**Iterate on feedback**	**Updated features**
	Extension improvements	**v0.2 release**
	API improvements	**API updates**
	Documentation updates	**Updated docs**
Week 36-37	**Continue beta testing**	**Ongoing support**
	Stability improvements	**Stable release**
	Performance tuning	**Optimized system**
	Prepare Chrome Web Store	**Store assets**

Sprint 10: Public Launch (Oct 27 - Dec 15)

Week	**Tasks**	**Deliverables**
------	**-------**	**----------**
Week 38	**Chrome Web Store submission**	**Submitted extension**
	Review response handling	**Updates if needed**
	Landing page	**thingsthatmatter.com**

	Documentation site	**docs.thingsthatmatter.com**
Week 39-40	**Marketing preparation**	**Blog posts, demos**
	Developer documentation	**SDK docs**
	API documentation	**API reference**
	Support documentation	**FAQ, troubleshooting**
Week 41-42	**Public launch**	**🚀 Live!**
	Monitor adoption	**Analytics**
	Support incoming users	**User support**
	Bug fixes	**Patch releases**
Week 43-44	**Post-launch stabilization**	**Stable v1.0**
	Retrospective	**Lessons learned**
	Phase 2 planning	**Phase 2 roadmap**
	Year-end wrap	**Annual report**

Milestone: 100% Complete

Exit Criteria: - [ ] 5-10 beta merchants onboarded - [ ] 1000+ products compiled - [ ] Extension in Chrome Web Store - [ ] Documentation complete - [ ] Public launch successful

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3. Milestone Summary

Milestone	Date	Completion	Key Deliverable
---	---	---	---
Contract Signed	Jan 21, 2026	0%	New company agreement
Phase 0 Complete	Mar 2, 2026	15%	Steganography validation
Foundation Complete	Apr 27, 2026	30%	Rust compiler MVP
Core Complete	Jul 20, 2026	50%	Full pipeline working
75% Milestone	Aug 17, 2026	75%	Browser extension functional
Beta Launch	Oct 26, 2026	90%	Merchants onboarded
Public Launch	Dec 15, 2026	100%	Chrome Web Store live

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4. Dependencies & Critical Path

4.1 Critical Path

```
Contract Sign → Phase 0 → Go/No-Go → FlatBuffers → Rust Compiler
                                                        ↓
                                                  CLIP Integration
                                                        ↓
                                                Steganography Engine
                                                        ↓
                                                 Shadow Index API
                                                        ↓
                                                Browser Extension
                                                        ↓
                                                  Beta Launch
                                                        ↓
                                                 Public Launch
```

4.2 Dependency Matrix

Task	Depends On	Blocks
------	----------	--------
Phase 0	**Contract signed**	**Go/No-Go decision**
FlatBuffers schema	**Phase 0 Go**	**Rust compiler**
Rust compiler	**FlatBuffers**	**Steganography**
CLIP integration	**API account**	**Rust compiler**
Ed25519 signing	**None**	**Compiler**
Steganography	**Rust compiler**	**Shadow Index**

Shadow Index API	Steganography	Browser extension
Browser extension	Shadow Index, WASM	Beta launch
Rust optimization	Rust compiler working	Performance tuning
Beta launch	Extension, merchants	Public launch

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5. Risk to Timeline

Risk	Probability	Impact	Mitigation	Schedule Impact
Phase 0 fails (<50% survival)	Medium	High	Pivot to matter://	+2 weeks
Cursor Pro limits slow dev	Medium	Medium	Batch work	+2-4 weeks
CLIP API pricing changes	Low	Medium	Self-host GPU	+4 weeks
Chrome rejects extension	Low	High	Firefox first	+4 weeks
Rust learning takes longer	Low	Low	C/C++ background accelerates learning	None
Merchant adoption slow	Medium	Medium	Developer focus	Shift strategy

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6. Resource Allocation

6.1 Time Allocation by Category

Category	Hours/Week	% of Time
---	---	---
Core development	20	67%
Learning (Rust)	3	10%
Documentation	2	7%
Testing	3	10%
Admin/meetings	2	6%
Total	**30**	**100%**

6.2 Monthly Budget Burn

Month	Amount	Cumulative	Key Expenses
---	---	---	---
Feb	R8,000	R8,000	Phase 0, setup
Mar	R6,500	R14,500	Development tools

Apr	**R6,000**	**R20,500**	**CLIP API**
May	**R6,000**	**R26,500**	**CLIP, storage**
Jun	**R6,300**	**R32,800**	**API hosting**
Jul	**R6,000**	**R38,800**	**Development**
Aug	**R6,200**	**R45,000**	**Extension dev**
Sep	**R6,000**	**R51,000**	**Testing**
Oct	**R6,000**	**R57,000**	**Beta support**
Nov	**R7,500**	**R64,500**	**Launch prep**
Dec	**R7,500**	**R72,000**	**Public launch**

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7. Progress Tracking

7.1 Weekly Checkpoints

Every Friday: - [ ] Code committed to repository - [ ] Progress update to CEO - [ ] Blockers identified - [ ] Next week planned

7.2 Monthly Demos

First Monday of each month: - [ ] Working demo of new features - [ ] Metrics review - [ ] Budget review - [ ] Timeline adjustment if needed

7.3 Milestone Reviews

At each milestone: - [ ] Deliverables verified - [ ] Quality assessment - [ ] Go/No-Go for next phase - [ ] Retrospective

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8. Success Criteria by Phase

Phase 0 Success (Feb 2026)

- [ ] ≥50% steganography survival rate - [ ] CLIP integration working - [ ] End-to-end prototype functional - [ ] Clear Go/No-Go decision

Foundation Success (Apr 2026)

- [ ] FlatBuffers schema finalized - [ ] Rust compiler producing valid .thing files - [ ] <500ms compile time - [ ] CLI tool operational

Core Success (Jul 2026)

- [ ] Steganography engine complete - [ ] Shadow Index API deployed - [ ] matter:// URIs generating - [ ] Rust port 30% complete

Distribution Success (Sep 2026)

- [ ] Browser extension functional - [ ] End-to-end demo working - [ ] WASM verification working - [ ] Ready for beta

Launch Success (Dec 2026)

- [ ] 5-10 merchants onboarded - [ ] 1000+ products compiled - [ ] Extension in Chrome Web Store - [ ] Public launch complete

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Document Version: 1.0 **Last Updated:** January 21, 2026 **Next Review:** Monthly